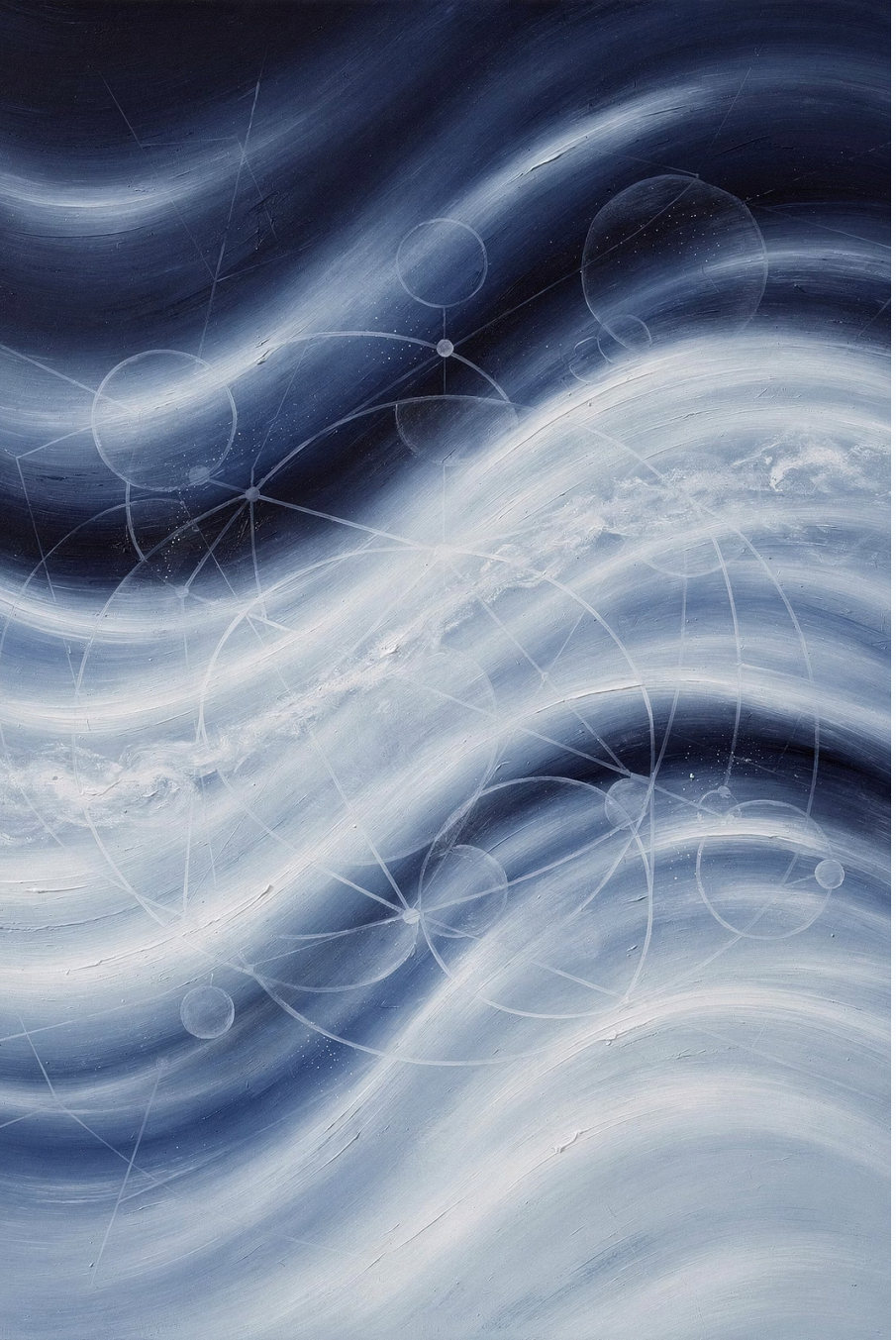


Augmenting Nuclear Engineering with AI

A Practical Roadmap for Capital Design Teams

Accelerating Engineering Workflows While Preserving Design Authority

---

Evan Parra — AI/ML Engineer
MS in Artificial Intelligence, Florida Atlantic University
Google Certified Machine Learning Engineer

Prepared for NextEra Energy Nuclear Engineering Leadership
March 2026

The Challenge Facing Nuclear Engineering Teams

Today's nuclear capital engineering environment is defined by compounding pressures that legacy workflows were never designed to handle at this scale or pace.

🌊 The Silver Tsunami

A generation of senior engineers is retiring, taking decades of plant-specific institutional knowledge with them. Replacement pipelines cannot keep pace.

⚡ Surging Baseload Demand

AI data centers are driving unprecedented electricity demand. Nuclear is the only 24/7 zero-carbon source that can reliably meet it — placing enormous pressure on existing fleets.

🏭 Extended Plant Life

Turkey Point Units 3 & 4 became the first plants to receive Subsequent License Renewal — operating well beyond original design life. Every modification carries greater scrutiny.

📄 Documentation Overload

Capital modifications require manual review of thousands of pages across UFSAR sections, vendor documents, and regulatory text — a process measured in days, not hours.

The Prime Directive

NON-NEGOTIABLE FOUNDATION

AI has zero design authority.

This is not a legal disclaimer buried in a footer — it is the architectural foundation of every tool, workflow, and deployment described in this roadmap. Before any use case, before any architecture decision, this principle must be technically enforced, not just acknowledged.

Cognitive Exoskeleton — Not a Replacement

AI augments the licensed engineer's capacity to process, retrieve, and draft. The engineer's judgment, stamp, and signature remain the only legally binding output.

Every Output is a Draft

Every AI-generated response — screening drafts, summaries, comparisons — carries a mandatory **"PRELIMINARY — REQUIRES LICENSED ENGINEER REVIEW"** banner. This is enforced at the application layer, not optional.

Bounded, Auditable, Traceable

No hallucinated citations. Every answer includes exact source document, section, and page number. Every query and response is logged to an immutable audit trail in BigQuery.

NextEra's AI Readiness: A Strong Foundation Already Exists

NextEra is not starting from zero. The transition from discriminative ML to generative AI builds directly on proven infrastructure and an established Google Cloud partnership.

Strategic Partnership (Dec 2025)

NextEra + Google Cloud partnership for gigawatt-scale data centers and enterprise-wide digital transformation — the same GCP infrastructure proposed in this roadmap.

- TimesFM 2.5 for load forecasting
- WeatherNext 2 for grid optimization
- Duane Arnold restart — nuclear powering AI data centers

Already Deployed at Nuclear Sites

- NLP for corrective action program trending
- Neural networks for procedure search and retrieval
- Clustering algorithms for preventive maintenance optimization

The Next Step

Generative AI — RAG pipelines, document Q&A, agentic workflows — is the natural evolution of these existing capabilities. The data infrastructure, the cloud partnership, and the organizational appetite are already in place.

Five High-Impact Use Cases

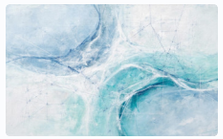
Each use case was selected based on three criteria: high engineer time burden, well-defined inputs and outputs, and clear regulatory boundaries that keep AI in an advisory role. Together, these five applications can recover thousands of engineering hours annually across a multi-unit site.

50.59 Screening

Days → Minutes. Automated UFSAR cross-reference with cited draft output.

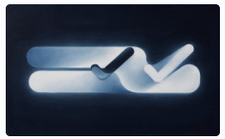
Package Summarization

500+ page packages distilled to key modifications, safety justifications, and open items.



Legacy Digitization

P&IDs and SLDs converted from scanned PDFs to searchable, traversable graph databases.



Vendor Bid Evaluation

RAG-powered spec compliance check — every technical deviation flagged automatically.

Knowledge Transfer

Plant-specific RAG chatbot — junior engineers query an always-available cited SME.

$1M+

Annual Savings

Per site from 50.59 screening automation alone

90%

Time Reduction

Screening and summarization tasks: days reduced to minutes

5

Use Cases

Bounded, auditable, and deployable on existing GCP infrastructure

USE CASE 1

10 CFR 50.59 Screening Automation

Current State

Engineers manually cross-reference a proposed modification description against multiple UFSAR chapters, design basis documents, and technical specifications. A single screening can consume **2–5 engineering days** — for a task that is largely information retrieval.

Industry Validation

- **Blue Wave AI 5059.ai** — deployed at operating utilities today
- **Nuclearn** — production-grade 50.59 AI screening tooling
- NRC has not objected to AI-assisted draft preparation with engineer review

AI-Assisted Workflow

Engineer submits a modification description → system ingests it, performs semantic search across chunked UFSAR → generates a **preliminary screening draft** with exact section citations and a recommended determination. Licensed engineer reviews, edits, and signs.

Quantified Impact

Screening Time

Days → ~30 minutes

Cost Savings

$1M+ per site annually

Output

Draft + citations. Engineer owns the determination.

⚠️ **PRELIMINARY — REQUIRES LICENSED ENGINEER REVIEW AND APPROVAL**

USE CASE 2

Engineering Package Summarization

Capital modification packages regularly exceed 500 pages — design change notices, engineering evaluations, vendor submittals, drawing markups, and open items scattered across sections. Engineers spend hours simply orienting themselves before technical review can begin.

1

Ingest Package

Upload 500+ page modification package to Cloud Storage. Document AI performs OCR and chunking.

2

Map-Reduce Summarization

Vertex AI processes chunks in parallel, extracts key content per section, then reduces to a coherent summary — the same strategy validated by Shell for CO_2 sequestration well reports.

3

Structured Output

Engineers receive: key modifications, impacted systems, safety justifications, open items — each with exact page references for rapid verification.

What Engineers Actually Get

- Executive summary of scope and safety basis
- Impacted systems and structures (with drawing references)
- Open items and action assignments with owners
- Safety justification excerpts with page citations

Platform

Google Cloud Vertex AI Jump Start Solutions for document summarization — production-ready, deployed on the same GCP infrastructure NextEra uses today.

⚠️ **PRELIMINARY — REQUIRES ENGINEER REVIEW**

USE CASE 3

Legacy Document Digitization

The Problem: Trapped Engineering Intelligence

Decades of P&IDs, cable schedules, single-line diagrams, and panel schedules exist only as scanned PDFs — unstructured, unsearchable, and requiring manual tracing to understand system relationships. When a senior engineer retires, so does their mental map of these documents.

AI-Driven Extraction

- **DiagramIQ** — purpose-built computer vision for P&ID symbol recognition and relationship extraction
- **Google Document AI** — high-accuracy OCR for handwritten markups and degraded scans
- Component relationships loaded into a **graph database** — engineers traverse upstream/downstream connections digitally

What Gets Unlocked

Searchable P&IDs

Query by tag number, component type, or system — instant results across all drawings

Auto-Generated Schedules

Cable schedules, load lists, and panel summaries auto-generated from SLD data

Impact Analysis

Modification impact traced automatically through graph — no more manual redlines

USE CASE 4

Vendor Bid Evaluation

Nuclear procurement is unforgiving: every technical deviation from an engineering specification — no matter how minor — must be identified, evaluated, and dispositioned. Manual evaluation of competing vendor submittals against complex specifications is time-consuming, inconsistent, and prone to evaluator fatigue.

Ingest Submittals

Upload vendor PDFs and engineering spec.

Deviation Report

Rank deviations with citations and severity flags.

RAG Comparison

Extract parameters, semantically match requirements.

Key Capabilities

- Semantic extraction of technical parameters from unstructured vendor submittals
- Parameter-by-parameter comparison against engineering specification requirements
- Automatic flagging of **every deviation** — nuclear procurement standard: zero tolerance for missed deviations
- Consistency enforcement across all evaluators — eliminates subjective variation

Why It Matters

Manual evaluation of three competing submittals against a 200-page spec can take a senior engineer a full week. AI reduces this to hours while improving traceability and defensibility in procurement audits.

> ⚠️ **All deviation flags require engineer review and engineering disposition**

USE CASE 5

Knowledge Transfer & Workforce Development

The Silver Tsunami Problem

When a 30-year nuclear engineer retires, they take with them an irreplaceable mental model of plant quirks, historical modifications, and tacit design basis knowledge that never made it into formal documentation. No onboarding program replaces that — but AI can help capture and transmit it.

Plant-Specific RAG Chatbot

- Ingests procedures, OEM manuals, design basis docs, CAP history, and engineering evaluations
- Junior engineers ask natural language questions and receive **cited answers** — exact document, section, and page
- Available 24/7, never too busy to answer, never impatient with a new engineer

Industry Precedents

EPRI TS-LLM

Structured Support/Refute tables with page citations for Technical Specification interpretation queries

NILLM Pilot

EPRI + INPO + NEI federated AI for cross-utility knowledge sharing — proven at the industry level

Strategic Value

This is the most durable long-term investment: capturing institutional knowledge before it walks out the door is a one-time opportunity.

Supporting Capability: NDE & Inspection Data Analysis

EPRI-VALIDATED

The Field Challenge

Ultrasonic testing of reactor vessel welds and safety-related piping generates massive volumes of raw waveform data. Traditionally, trained NDE Level III specialists review all data manually — a process measured in days per inspection campaign.

EPRI Field Trial Results

AI-assisted UT data review reduced full-dataset review time from **4 days to 4 hours** in EPRI field trials — a 24× improvement with no reduction in flaw detection accuracy.

How It Works

1 AI Filters Benign Data

Machine learning models trained on verified inspection datasets automatically classify normal indications — removing the noise that consumes most review time.

2 Anomalies Flagged for Expert Review

Ambiguous and potentially reportable indications are prioritized and surfaced to NDE specialists — human expertise focused where it matters most.

3 Direct Feed to Engineering Calculations

Flagged flaw dimensions are formatted for direct input to flaw growth and structural integrity calculations, closing the loop from inspection to engineering disposition.

Supporting Capability: Corrective Action Program Intelligence

GENERATIVE AI UPGRADE

NextEra already has baseline NLP capability for CAP trending. Generative AI is not a replacement — it is a significant step-change upgrade in insight depth, narrative clarity, and diagnostic utility.

Cross-Decade Trending

AI analyzes condition reports spanning 20+ years, surfacing latent failure patterns invisible to time-windowed manual trending. Identifies recurrence before it becomes a repeat significant condition adverse to quality.

Automated System Health Reports

Combines KPIs, failure trend data, and narrative insights into a coherent system health report — reducing the time a system engineer spends compiling inputs from days to under an hour.

Structured Diagnostic Game Plans

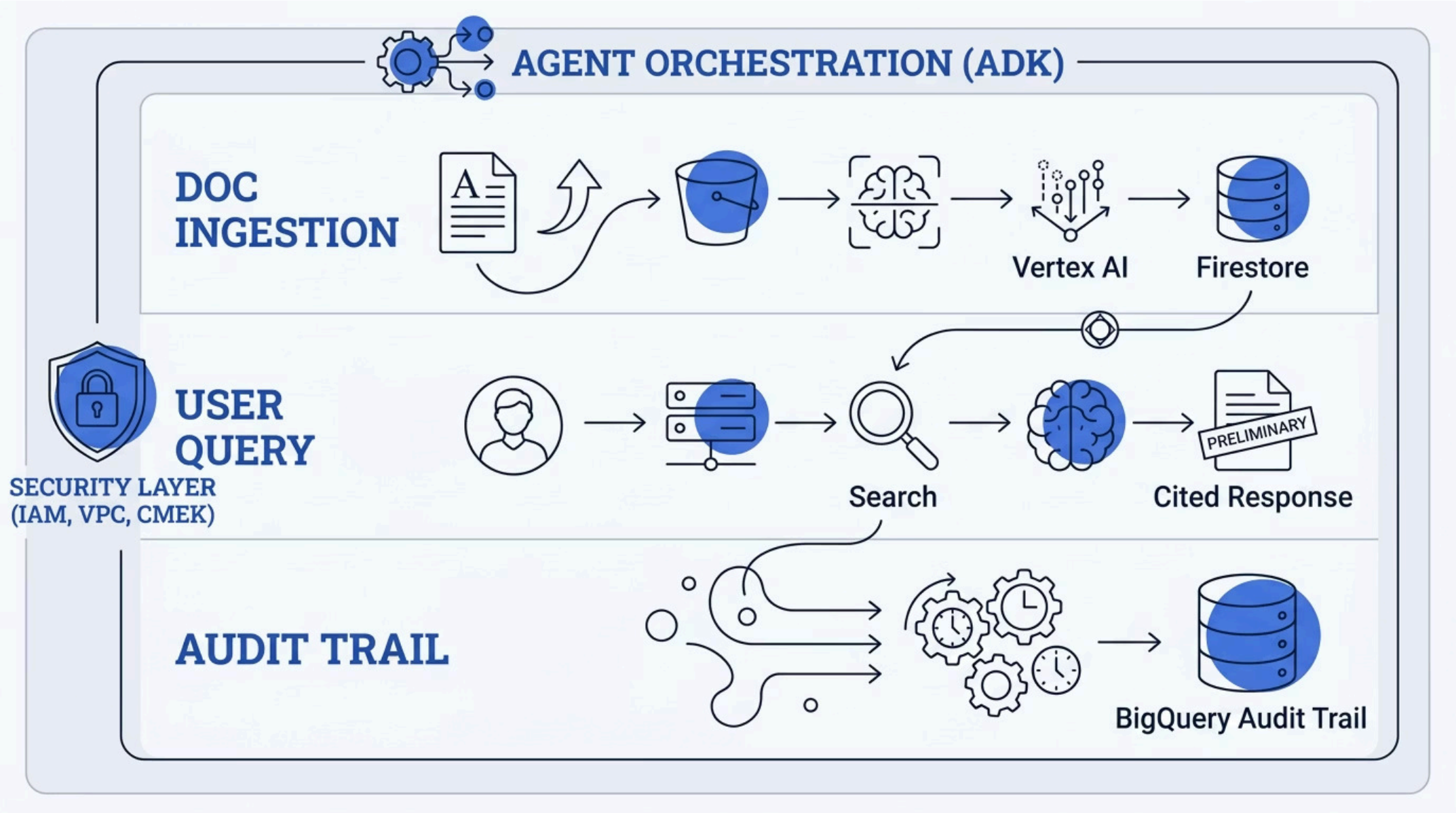
Nuclearn AtomAssist generates structured troubleshooting guides drawing on plant-specific CAP history and OEM documentation — giving craft and engineering staff a consistent diagnostic starting point.

Cross-System Correlation

Generative AI identifies condition reports in seemingly unrelated systems that share a common root cause — a capability that discriminative NLP models structurally cannot provide.

Architecture: Built on Google Cloud

Every component runs on the same GCP infrastructure NextEra committed to in its December 2025 strategic partnership — no new cloud vendors, no shadow IT, no architectural exceptions.



Document Ingestion

Cloud Storage → Document AI → Vertex AI Embeddings → Firestore Vector Search

Query & Generation

FastAPI (Cloud Run) → RAG Retrieval → Gemini → Cited Response

Audit & Compliance

Every query and response logged immutably to BigQuery — inspection-ready audit trail

Agent Orchestration

Google ADK for multi-step agentic workflows — 50.59 screening, vendor evaluation pipelines

Live Demo: Nuclear Engineering AI Assistant

PRODUCTION-GRADE | OPEN SOURCE

This is not a mockup or a concept deck. The application is built, running on GCP, and was constructed specifically to demonstrate nuclear engineering workflows — not adapted from a generic enterprise chatbot.

1

Document Q&A with Citation Table

Engineers query UFSAR sections, procedures, and design basis documents in natural language. Every response includes a structured citation table: document name, section, page number.

2

50.59 Screening Draft Generator

Input a modification description → receive a preliminary 50.59 screening draft with UFSAR cross-references populated, structured per 10 CFR 50.59(c)(2) criteria.

3

Mandatory Disclaimer Enforcement

Every generated output carries a persistent **"PRELIMINARY — REQUIRES LICENSED ENGINEER REVIEW"** banner enforced at the application layer. Non-dismissible, non-configurable.

Technical Stack

- Vertex AI + Gemini (generation)
- Firestore (vector search)
- Cloud Run / FastAPI (API layer)
- Google ADK (agent orchestration)
- BigQuery (audit logging)

Repository

github.com/DevDizzle/nuclear-eng-ai-assistant

Open source, documented, production-grade. Built for this team.

⚠️ **ALL OUTPUTS: PRELIMINARY — REQUIRES LICENSED ENGINEER REVIEW AND APPROVAL BEFORE USE**

Implementation Roadmap

A phased approach that delivers measurable value at each milestone — starting with the highest-impact, lowest-integration-complexity use cases and expanding as adoption matures and trust is established.

Phase 1 — Months 1–3

Foundation & Quick Wins

- Deploy document Q&A on existing UFSAR and regulatory docs
- Launch 50.59 screening assistant
- Pilot with 2–3 engineers on active modification projects
- Establish BigQuery audit baseline and usage metrics

Phase 2 — Months 4–6

Expand & Measure

- Add legacy P&ID and SLD digitization pipeline
- Deploy vendor bid evaluation RAG
- Expand pilot to full modification project team
- Publish adoption metrics and time-savings report to leadership

Phase 3 — Months 7–12

Scale & Institutionalize

- Scale to full engineering organization
- Build knowledge transfer RAG for workforce development
- Integrate generative AI upgrade to CAP trending
- Evaluate cross-site deployment for multi-unit portfolio

1 2 3

Phases

Structured milestones with measurable outcomes at each gate

Quick Win Coverage

Majority of high-impact use cases operational within 6 months

Month Horizon

Full deployment across engineering organization within one year

Strategic Imperatives

These are the four principles that must govern implementation — not aspirational guidelines, but enforceable design requirements that protect both the engineering organization and the public it serves.

1

Enshrine the Prime Directive

Technologically enforced guardrails at the application and infrastructure layer. Zero design authority is a system property, not a policy document. Mandatory review banners, role-based access, and workflow gates that prevent AI outputs from being treated as final engineering products.

2

Guarantee Data Sovereignty

Sensitive design basis documents are subject to **10 CFR Part 810 export controls**. On-premises or air-gapped GCP deployment for controlled documents. No training data leaves the boundary. Customer-managed encryption keys (CMEK) throughout. Full data residency compliance.

3

Prioritize High-Friction, Low-Risk Workflows

Start where engineers lose the most time with the least safety-critical direct output: 50.59 screening drafts, document summarization, vendor evaluation. Build trust through demonstrated accuracy before expanding to higher-stakes applications.

4

Demand Transparent AI

Every output must include exact citations — document name, section, page number — enabling engineer verification in under 60 seconds. Black-box outputs are disqualifying. Explainability is a procurement requirement, not a feature request.

Why This Comes Together With Me

EVAN PARRA — AI/ML ENGINEER

Production Systems on GCP

I have built and deployed production AI systems on the exact infrastructure proposed in this roadmap — not toy demos, but operational systems with real users, real data, and real SLAs.

→ ML & AI Engineering

Vertex AI Pipelines, BigQuery ML, Cloud Run, Gemini API, Google ADK — all used in production contexts, not just tutorials

→ RAG & Agentic Systems

Built and deployed RAG pipelines, multi-agent orchestration systems, and document Q&A applications from the ground up

→ Domain-Specific Application

Built the nuclear engineering AI assistant — github.com/DevDizzle/nuclear-eng-ai-assistant — specifically for this workflow and this audience

Credentials

- **MS in Artificial Intelligence** — Florida Atlantic University
- **Google Certified Machine Learning Engineer**

Local Commitment

From South Florida. Family in the area. Not a remote consultant parachuting in — I am ready to embed with the nuclear engineering team, learn the plant, and build tools that engineers actually use.

What I Bring

The rare combination of production AI engineering depth and the patience to understand a regulated, safety-critical environment on its own terms — not to disrupt it, but to augment it responsibly.

Thank You

Let's Build This Together

The tooling is ready. The infrastructure partnership is in place. The use cases are validated at peer utilities. What's needed now is an engineer who can bridge the gap between AI capability and nuclear engineering reality — and build the institutional trust to make it stick.

📧 Email

eraphaelparra@gmail.com

📞 Phone

(954) 548-7557

💻 GitHub

github.com/DevDizzle

🔗 LinkedIn

linkedin.com/in/evanparra

Case study also available at evanparra.ai — AI consulting capabilities for regulated industries